Functions: Composite & Inverse

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A vertical pole 6 m tall casts a shadow 9 m long. Find the angle of elevation of the sun, to the nearest tenth of a degree.

Final Answer: 33.7°. In the right triangle, opposite $= 6$, adjacent $= 9$, so the angle is $\tan^{-1}\left(\frac{2}{3}\right) \approx 33.7^\circ$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Solve $|2x - 4| < 2$. What is the upper bound of the solution interval? Show your complete algebraic work.

$-2 < 2x - 4 < 2 \Rightarrow 1 < x < 3$. Upper bound: $x = 3$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.

Final Answer: (1, 0). To find the intersection points, we set the two functions equal to each other: $f(x) = g(x)$ $x^2 - 4x + 3 = x - 1$ $x^2 - 5x + 4 = 0$ Factoring the quadratic equation: $(x - 1)(x - 4) = 0$ This gives us two possible x-values for the intersection points: $x = 1$ or $x = 4$. Using $g(x) = x - 1$ to find the corresponding y-values: For $x = 1$: $y = 1 - 1 = 0$. The intersection point is $(1, 0)$. For $x = 4$: $y = 4 - 1 = 3$. The intersection point is $(4, 3)$. Looking at the graph, one of the intersection po.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.

Final Answer: (3, 4). The given function is a quadratic of the form $f(x) = ax^2 + bx + c$, where $a = -1$, $b = 6$, and $c = -5$. The x-coordinate of the vertex of a parabola is given by $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{6}{2(-1)} = -\frac{6}{-2} = 3$. To find the y-coordinate of the vertex, substitute $x = 3$ into the function: $f(3) = -(3)^2 + 6(3) - 5 = -9 + 18 - 5 = 9 - 5 = 4$. Therefore, the coordinates of the vertex are (3, 4).

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown below. The graph passes through the point $(2, -3)$. Determine the value of k.

Final Answer: $k = 5$. The function is given by $f(x) = x^2 - 6x + k$. The graph of the function passes through the point $(2, -3)$. This means that when $x = 2$, $f(x) = -3$. We can substitute these values into the function equation: $-3 = (2)^2 - 6(2) + k$ $-3 = 4 - 12 + k$ $-3 = -8 + k$ To solve for k, add 8 to both sides of the equation: $-3 + 8 = k$ $5 = k$ Therefore, the value of k is 5.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(1, 0)$, $(5, 0)$, and $(0, -10)$. Find the value of a.

Final Answer: -2. The quadratic function has roots at $x = 1$ and $x = 5$. Therefore, it can be written in the factored form $f(x) = a(x - 1)(x - 5)$. We are given that the graph passes through the point $(0, -10)$. Substituting these coordinates into the factored form: $-10 = a(0 - 1)(0 - 5)$ $-10 = a(-1)(-5)$ $-10 = 5a$ Dividing both sides by 5, we get $a = -2$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = -2x + 6$. The graph of $y = f(x)$ is shown below. Determine the value of the y-intercept.

Final Answer: 6. The y-intercept of a function is the value of y when $x = 0$. For the function $f(x) = -2x + 6$, we substitute $x = 0$ into the equation: $f(0) = -2(0) + 6$ $f(0) = 0 + 6$ $f(0) = 6$ Therefore, the y-intercept is 6.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. What are the coordinates of the y-intercept? Show your complete algebraic work.

Final Answer: $(0, -3)$. The y-intercept of a function occurs when $x = 0$. Substituting $x = 0$ into the function $f(x) = -x^2 + 4x - 3$ gives $f(0) = -(0)^2 + 4(0) - 3 = -3$. Therefore, the coordinates of the y-intercept are $(0, -3)$.

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9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the set of values of the constant m for which the equation $|2x - 6| = mx + 2$ has exactly two distinct real solutions.

Final Answer: $-\frac{2}{3} < m < 2$. We consider the function $y = |2x - 6|$ and the line $y = mx + 2$. 1. **Analyze the modulus function:** The graph of $y = |2x - 6|$ consists of two linear rays meeting at the vertex $(3, 0)$: \[y = \begin{cases} 2x - 6, & \text{for } x \ge 3 \text{ (gradient } 2\text{)}, \\ -2x + 6, & \text{for } x -\frac{2}{3} \). 4. **Upper boundary (parallelism with the right ray):** The right ray has gradient 2. - If $m = 2$, the line $y = 2x + 2$ is parallel to $y = 2x - 6$ and never in.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the set of values of the constant k for which the equation $|x^2 - 4| = 2x + k$ has exactly four distinct real solutions.

Final Answer: $4 < k < 5$. To find the number of solutions to $|x^2 - 4| = 2x + k$, we consider the intersections of the curve $y = |x^2 - 4|$ and the line $y = 2x + k$. The curve is defined piecewise: - Inner arch: $y = 4 - x^2$ for $-2 \le x \le 2$ - Outer branches: $y = x^2 - 4$ for $x2$ 1. **Intersection with the inner arch ($-2 \le x \le 2$):** $4 - x^2 = 2x + k \Leftrightarrow x^2 + 2x + (k - 4) = 0$ For two distinct real intersections, the discriminant must be strictly positive: x The discriminant is.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The functions f and g are defined for all real values of x by $f(x) = 2x - 1$ and $g(x) = \frac{x+5}{3}$. Find the value of x such that $fg(x) = 9$.

Final Answer: $x = 10$. To find x such that $fg(x) = 9$, we can either determine the composite expression $fg(x)$ or solve sequentially from the outer function to the inner function. **Method 1 (Sequential Evaluation):** We have $f(g(x)) = 9$. Since $f(u) = 2u - 1$, setting $f(u) = 9$ gives: $2u - 1 = 9 \Rightarrow 2u = 10 \Rightarrow u = 5$. Thus, $g(x) = 5$. Now solve for x: $\frac{x+5}{3} = 5 \Rightarrow x + 5 = 15 \Rightarrow x = 10$. **Method 2 (Forming the Composite Function):**.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The quadratic equation $x^2 - (2k + 1)x + (k^2 + 2) = 0$ has real roots α and β. Given that $\alpha^2 + \beta^2 = 27$, find the value of the constant k.

Final Answer: 3. From the relations between roots and coefficients for the quadratic equation $x^2 - (2k + 1)x + (k^2 + 2) = 0$: $\alpha + \beta = 2k + 1$ and $\alpha\beta = k^2 + 2$ Expressing $\alpha^2 + \beta^2$ in terms of the sum and product of the roots: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ Substitute the known values: $(2k + 1)^2 - 2(k^2 + 2) = 27$ Expand and simplify: $(4k^2 + 4k + 1) - (2k^2 + 4) = 27$.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The line $y = mx - 1$ intersects the curve $y = x^2 + 3x + k$ at two distinct points for all real values of m. Find the range of values of the constant k.

Final Answer: $k < -1$. To find the points of intersection of the line and the curve, set their equations equal: $x^2 + 3x + k = mx - 1$ Rearranging into standard quadratic form $ax^2 + bx + c = 0$: $x^2 + (3 - m)x + (k + 1) = 0$ For the line and curve to intersect at two distinct points, the discriminant with respect to x, denoted Δ_x, must be strictly positive: $\Delta_x = (3 - m)^2 - 4(1)(k + 1) > 0$ Expanding and simplifying: $\Delta_x = m^2 - 6m + 9 - 4k - 4 = m^2 - 6m + (5 - 4k) > 0$ We are given tha.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

Final Answer: 2. For a rational function $f(x) = \frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x = 1$, we know that $x = 1$ is a root of the denominator $x^2 - bx + 2 = 0$. Therefore, $1^2 - b(1) + 2 = 0$, which simplifies to $1 - b + 2 = 0$, so $b = 3$. The denominator is $x^2 - 3x + 2 = (x - 1)(x - 2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

Final Answer: a = 5, b = 3. For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertica.

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Trigonometry**

The function f is defined for $x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Find an expression for $f^{-1}(x)$.

Final Answer: $f^{-1}(x) = \frac{3x+1}{x-2}$. To find the inverse function $f^{-1}(x)$, let $y = f(x)$: $y = \frac{2x+1}{x-3}$ Multiply both sides by $(x-3)$: $y(x-3) = 2x+1$ Expand the left-hand side: $xy - 3y = 2x+1$ Rearrange to collect all terms involving x on one side: $xy - 2x = 3y+1$ Factor out x: $x(y-2) = 3y+1$ Divide by $(y-2)$: $x = \frac{3y+1}{y-2}$ Replacing y with x gives: $f^{-1}(x) = \frac{3x+1}{x-2}, \quad x \neq 2$. Common m.

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Differentiation & Integration**

Consider the rational function $f(x) = \frac{ax^2+bx+c}{dx^2+ex+f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x\to\infty} f(x) = \frac{1}{2}$ and $\lim_{x\to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

Final Answer: $\frac{1}{2}$. For a rational function where the degree of the numerator is equal to the degree of the denominator, the limit as $x \to \infty$ is the ratio of the leading coefficients. Therefore, $\lim_{x\to\infty} f(x) = \frac{a}{d}$. We are given that this limit is $\frac{1}{2}$, so $\frac{a}{d} = \frac{1}{2}$. The condition $\lim_{x\to -1} f(x) = \infty$ implies that $x = -1$ is a vertical asymptote. This means that the denominator is zero when $x = -1$, and the numerator is.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability**

Consider the rational function $f(x) = \frac{x^2-9}{x-3}$. What is the value of $f(x)$ as x approaches 3? Show your complete algebraic work.

Final Answer: 6. The function is given by $f(x) = \frac{x^2-9}{x-3}$. We can factor the numerator as a difference of squares: $x^2 - 9 = (x-3)(x+3)$. Thus, for $x \neq 3$, we can simplify the function to $f(x) = \frac{(x-3)(x+3)}{x-3} = x+3$. To find the limit as x approaches 3, we can substitute 3 into the simplified expression: $\lim_{x\to 3} f(x) = \lim_{x\to 3} (x+3) = 3+3 = 6$. Although the function is undefined at $x = 3$, the limit exists and is 6. Therefore, as x approaches 3.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Circular Measure**

Consider the rational function $f(x) = \frac{x^2-4}{x-2}$. What is the value of the function as x approaches 2? Show your complete algebraic work.

Final Answer: 4. The given function is $f(x) = \frac{x^2-4}{x-2}$. We can factor the numerator as a difference of squares: $x^2 - 4 = (x-2)(x+2)$. Therefore, $f(x) = \frac{(x-2)(x+2)}{x-2}$. For $x \neq 2$, we can cancel the $(x-2)$ term, so $f(x) = x+2$. As x approaches 2, the value of $f(x)$ approaches $2+2 = 4$. Although the function is technically undefined at $x = 2$ due to division by zero, the limit as x approaches 2 exists.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The graph of the quadratic function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). Find the value of k.

Final Answer: 3. The graph of the function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). This means that when $x = 3$, $f(x) = 14$. We can substitute these values into the function's equation: $14 = 2(3)^2 - k(3) + 5$ $14 = 2(9) - 3k + 5$ $14 = 18 - 3k + 5$ $14 = 23 - 3k$ Now, we need to solve for k : $3k = 23 - 14$ $3k = 9$ $k = \frac{9}{3}$ $k = 3$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the natural domain and range of the function $f(x) = \sqrt{2x - 6}$.

a. **Domain:** $x \geq 3$**; Range:** $f(x) \geq 0$ ✓

b. Domain: $x > 3$; Range: $f(x) > 0$

c. Domain: $x \geq 0$; Range: $f(x) \geq -6$

d. Domain: $x \in \mathbb{R}$; Range: $f(x) \geq 0$

Radicand must be non-negative: $2x - 6 \geq 0 \Rightarrow x \geq 3$. Principal square root yields $f(x) \geq 0$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $g(x) = x^2 - 4x + 7$ is defined for $x \in \mathbb{R}$. What is the range of g?

a. $g(x) \geq 3$ ✓

b. $g(x) \geq 7$

c. $g(x) \geq -3$

d. $g(x) \in \mathbb{R}$

Complete the square: $g(x) = (x - 2)^2 + 3$. Minimum value is 3, so range is $g(x) \geq 3$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Explain why the function $h(x) = x^2 - 4$ for $x \in \mathbb{R}$ does not have an inverse.

a. It is a many-to-one function on $\mathbb{R}$ ✓

b. Its domain does not include negative numbers

c. Its range is bounded below by -4

d. It is not continuous at $x = 0$

An inverse requires a one-to-one function. Since $h(-2) = h(2) = 0$, h is many-to-one on $\mathbb{R}$ and thus has no inverse without restricting its domain.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find the expression for $fg(x)$.

a. $2x^2 + 1$ ✓

b. $4x^2 + 12x + 8$

c. $2x^2 + 5$

d. $2x^2 - 2$

$fg(x) = f(g(x)) = f(x^2 - 1) = 2(x^2 - 1) + 3 = 2x^2 - 2 + 3 = 2x^2 + 1$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $f(x) = \frac{1}{x+2}$ for $x \neq -2$ and $g(x) = 3x - 1$, solve the equation $gf(x) = 1$.

a. $x = 1/2$ ✓

b. $x = -1$

c. $x = 2$

d. $x = 0$

$gf(x) = g(\frac{1}{x+2}) = 3(\frac{1}{x+2}) - 1 = 1 \Rightarrow \frac{3}{x+2} = 2 \Rightarrow 2x + 4 = 3 \Rightarrow 2x = -1 \Rightarrow x = -1/2$? Wait: $3/(x+2) = 2 \Rightarrow x + 2 = 1.5 \Rightarrow x = -0.5$. Let's check choices: choice A: $x = -1/2$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $f(x) = 3x - 4$, find the expression for $ff(x)$.

a. $9x - 16$ ✓

b. $9x - 12$

c. $6x - 8$

d. $9x^2 - 24x + 16$

$ff(x) = f(3x - 4) = 3(3x - 4) - 4 = 9x - 12 - 4 = 9x - 16$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x-5}{x+1}$ $(x \neq -1)$.

a. $f^{-1}(x) = \frac{x+5}{2-x}$ ✓

b. $f^{-1}(x) = \frac{x-5}{2-x}$

c. $f^{-1}(x) = \frac{2x+1}{x-5}$

d. $f^{-1}(x) = \frac{x+1}{2x-5}$

Let $y = \frac{2x-5}{x+1} \Rightarrow y(x+1) = 2x-5 \Rightarrow yx+y = 2x-5 \Rightarrow yx-2x = -y-5 \Rightarrow x(2-y) = y+5 \Rightarrow x = \frac{y+5}{2-y}$. Thus $f^{-1}(x) = \frac{x+5}{2-x}$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $f(x) = x^2 - 6x + 10$ is defined for $x \geq 3$. Find the expression for $f^{-1}(x)$.

a. $f^{-1}(x) = 3 + \sqrt{x-1}$ **for** $x \geq 1$ ✓

b. $f^{-1}(x) = 3 - \sqrt{x-1}$ for $x \geq 1$

c. $f^{-1}(x) = \sqrt{x+6} - 3$

d. $f^{-1}(x) = 3 + \sqrt{x+1}$

$y = (x-3)^2 + 1 \Rightarrow (x-3)^2 = y-1 \Rightarrow x-3 = \sqrt{y-1}$ (since $x \geq 3$) $\Rightarrow x = 3 + \sqrt{y-1}$. Thus $f^{-1}(x) = 3 + \sqrt{x-1}$ for $x \geq 1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A function $g(x) = \frac{ax+b}{cx+d}$ is a self-inverse function ($g(x) = g^{-1}(x)$). Which condition guarantees this property for all valid x?

a. $a + d = 0$ ✓

b. $ad - bc = 1$

c. $a = d$

d. $b = c$

Inverting $y = \frac{ax+b}{cx+d}$ yields $x = \frac{-dy+b}{cy-a}$. For $g(x) = g^{-1}(x)$, $-d = a$, which means $a + d = 0$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the geometric relationship between the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$?

a. **Reflection in the line** $y = x$ ✓

b. Reflection in the x-axis

c. Reflection in the y-axis

d. Rotation of 180° about the origin

Since the coordinates (x, y) on f become (y, x) on f^{-1}, the two graphs are reflections of each other across the diagonal line $y = x$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the equation $f(x) = f^{-1}(x)$ where $f(x) = 2x - 3$.

a. $x = 3$ ✓

b. $x = 1$

c. $x = -3$

d. $x = 2$

Intersections of f and f^{-1} for increasing functions lie on $y = x$. Solve $2x - 3 = x \Rightarrow x = 3$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function $f(x) = 2x^2 - 8x + c$ has domain $x \geq k$. Find the least value of k for which f has an inverse.

a. $k = 2$ ✓

b. $k = 4$

c. $k = 0$

d. $k = -2$

The parabola has vertex at $x = -(-8)/(4) = 2$. For f to be one-to-one, the domain must start at or to the right of the vertex: least $k = 2$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What are the values of x that satisfy the equation $|3x-2| = 2|x+1|$?

a. $x = 0$ **and** $x = 4$ ✓

b. $x = 0$ and $x = -4$

c. $x = \frac{1}{5}$ and $x = 4$

d. $x = \frac{1}{4}$ and $x = \frac{3}{2}$

To solve the modulus equation $|3x-2| = 2|x+1|$, we can either consider the linear cases or square both sides. **Method 1: Considering linear cases** The equation $|A| = |B|$ (where $B = 2(x+1)$) is equivalent to $A = B$ or $A = -B$: - Case 1: $3x-2 = 2(x+1) \Rightarrow 3x-2 = 2x+2 \Rightarrow x = 4$ - Case 2: $3x-2 = -2(x+1) \Rightarrow 3x-2 = -2x-2 \Rightarrow 5x = 0 \Rightarrow x = 0$ **Method 2: Squaring both sides** Since both sides are non-negative, we can square both sides.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The function f is defined for $x \in \mathbb{R}$ by $f(x) = x^2 - 2kx + (3k+10)$, where k is a real constant, and the equation $f(x) = 0$ has no real roots. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x-k|$. Find the set of values of k for which the minimum value of the composite function $h(x) = f(g(x))$ is strictly greater than 4.

a. $-2 < k < \frac{3+\sqrt{33}}{2}$ ✓

b. $\frac{3-\sqrt{33}}{2} < k < \frac{3+\sqrt{33}}{2}$

c. $-2 < k < 5$

d. $0 \le k < \frac{3+\sqrt{33}}{2}$

We solve the problem in three systematic stages: 1. **Condition from the Discriminant of $f(x) = 0$:** Since $f(x) = 0$ has no real roots, its discriminant Δ must be strictly negative: x Combining with $k < 0$, this yields $-2 < k < 0$. 3. **Combining the Conditions:** Combining Case 1 and Case 2 gives: $(-2, 0) \cup \left[0, \frac{3+\sqrt{33}}{2}\right) = -2 < k < \frac{3+\sqrt{33}}{2}$. Since $\frac{3+\sqrt{33}}{2} \approx 4.372 < 5$, this entire interval lies within the initial constrai.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x+1}{x-2}$ ✓

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

To find the inverse function $f^{-1}(x)$, set $y = f(x)$ and rearrange to express x in terms of y: $y = \frac{2x+1}{x-3}$ Multiply both sides by $x - 3$: $y(x-3) = 2x + 1$ $yx - 3y = 2x + 1$ Collect all terms containing x on one side and constant/independent terms on the other: $yx - 2x = 3y + 1$ Factor out x: $x(y-2) = 3y + 1$ Divide by $y - 2$: $x = \frac{3y+1}{y-2}$ Interchanging the variables gives: $f^{-1}(x) = \frac{3x+1}{x-2}$ Common Misconceptions: - Choice.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function f is defined by $f(x) = \frac{2x-1}{x+3}$, for $x \neq -3$. Which of the following expressions represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+1}{2-x}, \quad x \neq 2$ ✓

b. $f^{-1}(x) = \frac{x+3}{2x-1}, \quad x \neq \frac{1}{2}$

c. $f^{-1}(x) = \frac{3x-1}{x-2}, \quad x \neq 2$

d. $f^{-1}(x) = \frac{2x+1}{x-3}, \quad x \neq 3$

To find the inverse function $f^{-1}(x)$: 1. Set $y = f(x)$: $y = \frac{2x-1}{x+3}$ 2. Multiply both sides by the denominator $x + 3$: $y(x+3) = 2x - 1 \Rightarrow xy + 3y = 2x - 1$ 3. Collect all terms containing x on one side and constant/y terms on the other: $xy - 2x = -3y - 1$ 4. Factor out x: $x(y-2) = -(3y+1)$ 5. Solve for x: $x = \frac{-(3y+1)}{y-2} = \frac{3y+1}{2-y}$ 6. Express in terms of x: $f^{-1}(x) = \frac{3x+1}{2-x}, \quad x \neq 2$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $xe2$. Find the inverse function $f^{-1}(x)$.

a. $f^{-1}(x) = \frac{3x-2}{x+1}$

b. $f^{-1}(x) = \frac{2x-1}{x-3}$

c. $f^{-1}(x) = \frac{x-2}{3x+1}$

d. $f^{-1}(x) = \frac{2x+1}{x-3} \quad (xe3)$ ✓

Let $y = \frac{3x+1}{x-2} \Rightarrow y(x-2) = 3x+1 \Rightarrow yx-2y = 3x+1 \Rightarrow x(y-3) = 2y+1 \Rightarrow x = \frac{2y+1}{y-3}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-3}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

For $0 < x < 1$, simplify $\cos(\sin^{-1}(x))$ as an algebraic expression in x.

a. $\sqrt{1-x^2}$ ✓

b. $\frac{1}{\sqrt{1-x^2}}$

c. $1-x^2$

d. $\sqrt{x^2-1}$

Let $\theta = \sin^{-1}(x)$, so $\sin\theta = x$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, where $\cos\theta \geq 0$. Then $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-x^2}$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Circular Measure

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $(-\frac{\pi}{2}, \frac{\pi}{2})$, range $(-\infty, \infty)$

d. **Domain** $(-\infty, \infty)$**, range** $(-\frac{\pi}{2}, \frac{\pi}{2})$ ✓

$\tan^{-1}(x)$ has domain $(-\infty, \infty)$ (restricted input) and range $(-\frac{\pi}{2}, \frac{\pi}{2})$ (restricted principal output).

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

c. **Domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$** ✓

d. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (restricted principal output).

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Series & Binomial Theorem

A 13 m ladder leans against a vertical wall and reaches 12 m up the wall. Find the angle the ladder makes with the ground, to the nearest tenth of a degree.

a. 42.7°

b. 22.6°

c. 72.4°

d. 67.4° ✓

In the right triangle, opposite $= 12$, hypotenuse $= 13$, so the angle is $\sin^{-1}(\frac{12}{13}) \approx 67.4^\circ$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Kinematics

A kite string 50 m long is pulled taut, and the kite is 32 m above the hand holding it. Find the angle of elevation of the string, to the nearest tenth of a degree.

a. 50.2°

b. 39.8° ✓

c. 32.6°

d. 44.8°

In the right triangle, opposite $= 32$, hypotenuse $= 50$, so the angle is $\sin^{-1}(\frac{16}{25}) \approx 39.8^\circ$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 12.01 Forces & Equilibrium

The function $f(x) = (x + 6)^2 - 3$ has domain $x \geq -6$. State the range of f.

a. $f(x) \leq -3$

b. $f(x) \geq -3$ ✓

c. $f(x) \geq -6$

d. $x \geq -6$

Since $x \geq -6$, the minimum of f occurs at $x = -6$, giving $f(-6) = -3$, and f increases thereafter. Range: $f(x) \geq -3$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 13.01 Newton's Laws

A 15 m rope is tied to the top of a pole and staked to level ground 9 m from the base of the pole. Find the angle the rope makes with the ground, to the nearest tenth of a degree.

a. 36.9°

b. 31.0°

c. 58.1°

d. 53.1° ✓

In the right triangle, adjacent $= 9$, hypotenuse $= 15$, so the angle is $\cos^{-1}\left(\frac{3}{5}\right) \approx 53.1^\circ$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 14.01 Energy, Work & Power

A ramp rises 3 m over a horizontal distance of 20 m. Find its angle of inclination, to the nearest tenth of a degree.

a. 8.5° ✓

b. 81.5°

c. 13.5°

d. 17.1°

In the right triangle, opposite $= 3$, adjacent $= 20$, so the angle is $\tan^{-1}\left(\frac{3}{20}\right) \approx 8.5^\circ$.

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